

WEEK 1 • DAY 1 ASSIGNMENT

 Generative AI Models & Transformers

A clean comparison of modern AI model providers, public/private models, and the core transformer workflow.

LLM Providers

Token Cost

Context Window

Transformers

1 10 Different Model Providers and Their Models

S.No	Provider	Model	Release Date	Cutoff Date	Cost of Tokens	Context Window	Public/Private	When to Use This Model
1	OpenAI	GPT-5.5	Apr 2026	Nov 2025	\$5-30 / 1M	1M tokens	Public	Reasoning, coding, AI agents
2	Anthropic	Claude Opus 4.7	Apr 2026	Jan 2026	\$5-25 / 1M	1M tokens	Public	Long workflows, research
3	Google	Gemini 3.1 Pro	Feb 2026	Jan 2025	\$2-12 / 1M	1M tokens	Public	Multimodal AI, large documents
4	DeepSeek	DeepSeek V4	Apr 2026	May 2025	\$0.44-0.88 / 1M	1M tokens	Public	Cheap reasoning, coding
5	Meta	Llama 3.1 (70B)	Jul 2024	Dec 2023	Free/Open	128K tokens	Public/Open	Research, fine-tuning
6	Mistral AI	Mistral Large 3	Dec 2025	Dec 2025	\$0.50-1.50 / 1M	256K tokens	Public	Enterprise AI
7	xAI	Grok 4.3	Apr 2026	Dec 2025	\$1.25-2.50 / 1M	1M tokens	Private	Real-time social data
8	Alibaba	Qwen 3.5 Flash	Feb 2026	Jan 2026	\$0.10-0.40 / 1M	1M tokens	Public	Fast scalable AI apps
9	Databricks	DBRX	Mar 2024	2024	Open Source	32K tokens	Public/Open	Enterprise open-source AI
10	NVIDIA	Nemotron 3 Nano	Dec 2026	Nov 2025	\$0.05-0.24 / 1M	1M tokens	Public	RAG and document analysis

Note: Model prices, availability, and release dates can change. For final academic submission, verify details from official provider pages.

 Easy Notes to Remember Models

Model	Easy Memory Trick
GPT	Smart reasoning AI 
Claude	Calm research assistant 

 Public Models vs Private Models

Public Models

- Open-source or openly accessible
- Free or customizable

Private Models

- Closed-source
- Usually paid APIs
- Hidden architecture

Model	Easy Memory Trick	
Gemini	Text + image AI	👤🖼️
Llama	Free open-source AI	🦙
Mistral	Fast enterprise AI	🦋
DeepSeek	Cheap powerful AI	💰
Grok	Live internet AI	⚡
Qwen	Fast and low-cost AI	🚀
DBRX	Enterprise research AI	🖋️
Nemotron	Document understanding AI	📄

- More transparent
- Example: Llama

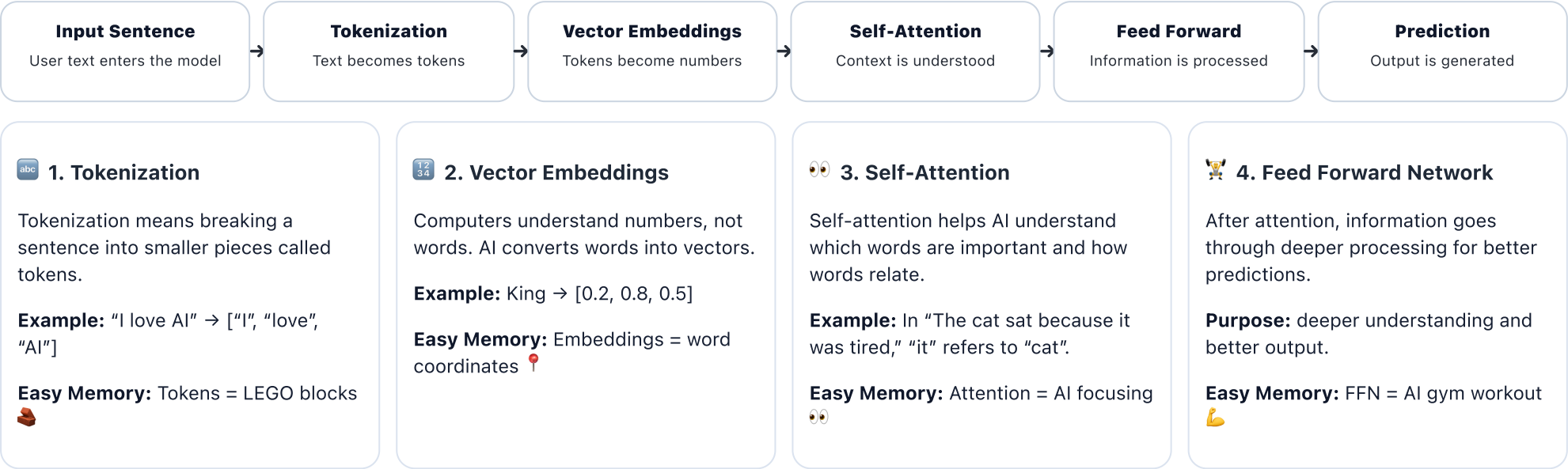
- Example: GPT

2 Understanding How Transformers Work

🤖 What is a Transformer?

A Transformer is the main architecture used in modern AI systems like ChatGPT, Gemini, Claude, and Llama. Transformers help AI understand language, learn context, and generate accurate responses.

🔄 Transformer Workflow



✅ Advantages of Public Models

Advantage	Meaning
Cost Efficient	Saves money
Customizable	Can modify model
Transparent	Easy to study

🏁 Conclusion

Modern AI systems are built using transformer architecture. Important concepts include tokenization, vector embeddings, self-attention, and feed forward networks.

Different AI companies create models for coding, reasoning, multimodal AI, enterprise applications, and research. Public models like Llama and Qwen are

Advantage	Meaning
Community Support	Developers improve models
No Vendor Lock-in	More flexibility

useful for learning and customization, while private models like GPT and Claude provide advanced enterprise-grade performance.